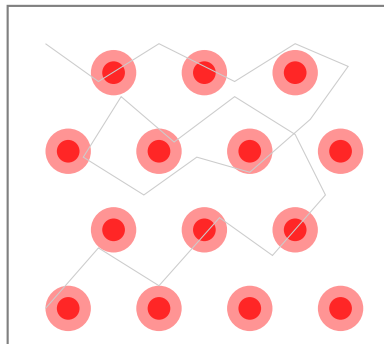


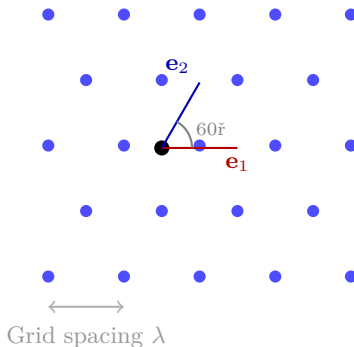
Grid Cell Hexagonal Firing Pattern

(A) Spatial firing fields

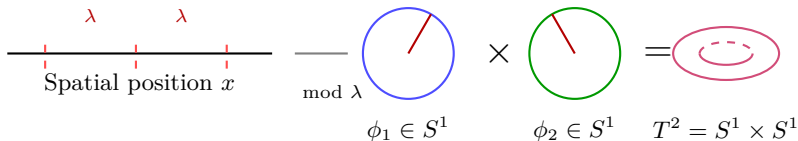


Position in arena

(B) Hexagonal lattice structure



(C) Phase wrapping produces toroidal topology



Periodic spatial coding in two directions produces two cyclic phase variables. Their joint state space is the two-torus T^2 .